Cost_change_source_cluster_documentation

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R Markdown

Documentation for R script: Cost_change_source_cluster.r which is called by “LASTING_cluster_w_masterinput.r” and is used to calculate cost, environment and social pillar impact for a given dietary type and food source (food at home vs food away from home). It will calculate population level and per capita results, and output the full sims, as well as summary stats, into separate output files.

Don’t be confused by the name of the file or some of the object names within this script. This is adapted from code that was originally for calculating cost change only, hence the “econ” specific naming convnetiosn peppered throughout.

First, read in functions used to calculate impacts.

```
#Cost_change_source.r
#setwd("C:\\Users\\fcudhe01\\Box Sync\\new life\\LASTING\\Code")
setwd("/cluster/tufts/lasting/shared/LASTING/Code")
source("cost_change_functions.r")
```

Define working directory and output location.

```
#file_location = "C:\\Users\\fcudhe01\\Box\\lasting_aim_3\\data\\in\\example_cra"
#file location already set in "LASTING.r"
setwd(file_location)

output_location = paste("/cluster/tufts/lasting/shared/model_development/data/outputs/cost_env/",
                        costs_output_loc.vec[k], sep="")
```

Read in relevant inputs (population draws, costs and impact factors of dietary factors) and do some minor data processing. Remember, the “impact factors” for cost is the cost for that dietary factor. We are not speculating how much costs for each dietary factor change as intake changes, just how much total spend on food would change (assuming costs stay the same).

```
#pop.sims<-read.csv("population.sims.csv", fileEncoding = 'UTF-8-BOM')
pop.sims<-read.csv("observed.pop.draws.csv", fileEncoding = 'UTF-8-BOM')
#pop.sims$subgroup_id<-pop.sims$subgroup

cost.inputs<-costs_inputs.vec[[k]]
merged<-merge(cost.inputs, pop.sims, by="subgroup_id")
merged$Sex<-merged$Sex.y
merged$population <- merged$population.y
```

Create list objects to store outputs. Each element of the lists corresponds to a specific subgroup. There are 31 lists in total, corresponding so some combination of 1. what we are looking at (impact (as in, impact of change), substitution impact, combined impact, population, delta (aka, change in intake), current intake impact, counterfactual intake impact), and 2. level of disaggregation (total, consumed, unconsumed, inedible, wasted). Hopefully, what each list corresponds to is clear from name.

```
################
list.of.impact.sims.total<-list()
list.of.substitution.impact.sims.total<-list()
list.of.combined.impact.sims.total<-list()
list.of.population.sims<-list()
list.of.delta.sims.total<-list()

list.of.impact.sims.consumed<-list()
list.of.substitution.impact.sims.consumed<-list()
list.of.combined.impact.sims.consumed<-list()
list.of.delta.sims.consumed<-list()

list.of.impact.sims.unconsumed<-list()
list.of.substitution.impact.sims.unconsumed<-list()
list.of.combined.impact.sims.unconsumed<-list()
list.of.delta.sims.unconsumed<-list()

list.of.current.intake.impact.sims.total<-list()
list.of.CF.intake.impact.sims.total<-list()
list.of.current.intake.impact.sims.consumed<-list()
list.of.CF.intake.impact.sims.consumed<-list()
list.of.current.intake.impact.sims.unconsumed<-list()
list.of.CF.intake.impact.sims.unconsumed<-list()

list.of.impact.sims.inedible<-list()
list.of.substitution.impact.sims.inedible<-list()
list.of.combined.impact.sims.inedible<-list()
list.of.delta.sims.inedible<-list()

list.of.impact.sims.wasted<-list()
list.of.substitution.impact.sims.wasted<-list()
list.of.combined.impact.sims.wasted<-list()
list.of.delta.sims.wasted<-list()

list.of.current.intake.impact.sims.inedible<-list()
list.of.CF.intake.impact.sims.inedible<-list()
list.of.current.intake.impact.sims.wasted<-list()
list.of.CF.intake.impact.sims.wasted<-list()
```

Rename some variables.

```
n.sims<-nsim1
impact.factor.names<-econ.outcomes.vec
```

Start for loop. We are going to be traversing the "merged" file row by row and calculating impacts. Each row should correspond to a subgroup and dietary factor.

```
for(i in 1:dim(merged)[1]) ##for each row in the input file (corresponding to fcid for a given subgroup
{
```

Still in loop: Extract label variables and put them into their own data frames (to be merged later with the results).

```
print(i)
vars.to.keep.for.df.a<-c("Foodgroup", "sex_gp", "race_gp", "age_gp",
                         "subgroup_id", "population",
                         "Mean_Intake", "SE_Intake", "CF_Mean_Intake", "CF_SE_Intake", "Intake_unit",

                         # "CED_unit", "GHG_unit", "WATER_unit","BLUEWATER_unit", "Food_price_unit",
                         # "CED_to_intake_conversion",
                         # "GHG_to_intake_conversion",
                         # "WATER_to_intake_conversion",
                         # "BLUEWATER_to_intake_conversion",
                         # "Food_price_to_intake_conversion",
                         # "FL_to_intake_conversion",

                         paste0(all.outcomes.vec, "_unit"),
                         paste0(all.outcomes.vec, "_to_intake_conversion"))

df.a<-merged[i, vars.to.keep.for.df.a]
```

Still in loop: Here, we are just creating a data frame that repeats df.a for however many outcomes we are looking at (six, in our case). We are going to have results for each pair of dietary factor and outcome.

```
df.aa<-df.a[rep(seq_len(nrow(df.a)), each = length(impact.factor.names)),]
```

Still in loop: For the sake of readability of code, we extract various relevant inputs from merged[i,] and give them their own object with own name.

```
current.mean<-merged[i,"Mean_Intake"]
current.se<-merged[i,"SE_Intake"]
counterfactual.mean<-merged[i,"CF_Mean_Intake"]
counterfactual.se<-merged[i,"CF_SE_Intake"]
impact.factor.means<-merged[i,econ.outcomes.vec.mn]
impact.factor.ses<-merged[i,econ.outcomes.vec.se]
#RRunit<-merged[i, "price_to_intake_conversion"]

# idk why this isn't working
# RRunit<-merged[i, c("CED_to_intake_conversion", "GHG_to_intake_conversion",
#                      "WATER_to_intake_conversion", "BLUEWATER_to_intake_conversion",
#                      "Food_price_to_intake_conversion", "FL_to_intake_conversion")]

# new
RRunit <- merged %>% select(c(paste0(impact.factor.names, "_to_intake_conversion"))) %>%
  slice(1)


pop.sims.names<-paste("X", c(1:n.sims), sep="")
population.sims<-matrix(rep(x=as.numeric(merged[i, pop.sims.names]), times=length(impact.factor.names
#population.sims<-as.numeric(merged[i, pop.sims.names])
#population<-merged[i, "population"]
#population.se<-merged[i, "population.se"]

substitution.impact.factor.means<-merged[i,econ.outcomes.substitution.vec.mn]
substitution.impact.factor.ses<-merged[i,econ.outcomes.substitution.vec.se]
substitution_unit<-merged[i,"substitution_unit"]

current_inedible_p<-merged[i,econ.current.inedible.p.mn]
current_inedible_p_se<-merged[i,econ.current.inedible.p.se]
current_foodwaste_p<-merged[i,econ.current.foodwaste.p.mn]
current_foodwaste_p_se<-merged[i,econ.current.foodwaste.p.se]
counterfactual_inedible_p<-merged[i,econ.counterfactual.inedible.p.mn]
counterfactual_inedible_p_se<-merged[i,econ.counterfactual.inedible.p.se]
counterfactual_foodwaste_p<-merged[i,econ.counterfactual.foodwaste.p.mn]
counterfactual_foodwaste_p_se<-merged[i,econ.counterfactual.foodwaste.p.se]
```

Still in loop: Take the above inputs and feed it into function that simulates the impact (simulate.impact). The resulting “b” is a list of 31 matrices with simulation results that correspond to the 31 lists we made earlier.

```
b<-simulate.impact(current.mean=current.mean, current.se=current.se, counterfactual.mean=counterfactu
                   counterfactual.se=counterfactual.se, impact.factor.names=impact.factor.names,
                   impact.factor.means=as.numeric(impact.factor.means), impact.factor.ses=as.numeric(
                   RRunit=RRunit,
                   substitution.impact.factor.means=as.numeric(substitution.impact.factor.means),
                   substitution.impact.factor.ses=as.numeric(substitution.impact.factor.ses),
                   substitution_unit=substitution_unit,
                   current_inedible_p=as.numeric(current_inedible_p),
                   current_inedible_p_se=as.numeric(current_inedible_p_se),
                   current_foodwaste_p=as.numeric(current_foodwaste_p),
                   current_foodwaste_p_se=as.numeric(current_foodwaste_p_se),
                   counterfactual_inedible_p=as.numeric(counterfactual_inedible_p),
                   counterfactual_inedible_p_se=as.numeric(counterfactual_inedible_p_se),
                   counterfactual_foodwaste_p=as.numeric(counterfactual_foodwaste_p),
                   counterfactual_foodwaste_p_se=as.numeric(counterfactual_foodwaste_p_se),
                   n.sims=n.sims,
                   population.sims=population.sims)
```

Still in loop: Extract results from “b” (which itself is a list of 31 matrices) # and create data frames with proper labels.

```
df.b1<-data.frame(outcome=rownames(b[[1]]), b[[1]])
df.b2<-data.frame(outcome=rownames(b[[2]]), b[[2]])
df.b3<-data.frame(outcome=rownames(b[[3]]), b[[3]])
df.b4<-data.frame(outcome=rownames(b[[4]]), b[[4]])
df.b5<-data.frame(outcome=rownames(b[[5]]), b[[5]])

df.b6<-data.frame(outcome=rownames(b[[6]]), b[[6]])
df.b7<-data.frame(outcome=rownames(b[[7]]), b[[7]])
df.b8<-data.frame(outcome=rownames(b[[8]]), b[[8]])
df.b9<-data.frame(outcome=rownames(b[[9]]), b[[9]])

df.b10<-data.frame(outcome=rownames(b[[10]]), b[[10]])
df.b11<-data.frame(outcome=rownames(b[[11]]), b[[11]])
df.b12<-data.frame(outcome=rownames(b[[12]]), b[[12]])
df.b13<-data.frame(outcome=rownames(b[[13]]), b[[13]])

df.b14<-data.frame(outcome=rownames(b[[14]]), b[[14]])
df.b15<-data.frame(outcome=rownames(b[[15]]), b[[15]])
df.b16<-data.frame(outcome=rownames(b[[16]]), b[[16]])
df.b17<-data.frame(outcome=rownames(b[[17]]), b[[17]])
df.b18<-data.frame(outcome=rownames(b[[18]]), b[[18]])
df.b19<-data.frame(outcome=rownames(b[[19]]), b[[19]])

df.b20<-data.frame(outcome=rownames(b[[20]]), b[[20]])
df.b21<-data.frame(outcome=rownames(b[[21]]), b[[21]])
df.b22<-data.frame(outcome=rownames(b[[22]]), b[[22]])
df.b23<-data.frame(outcome=rownames(b[[23]]), b[[23]])

df.b24<-data.frame(outcome=rownames(b[[24]]), b[[24]])
df.b25<-data.frame(outcome=rownames(b[[25]]), b[[25]])
df.b26<-data.frame(outcome=rownames(b[[26]]), b[[26]])
df.b27<-data.frame(outcome=rownames(b[[27]]), b[[27]])

df.b28<-data.frame(outcome=rownames(b[[28]]), b[[28]])
df.b29<-data.frame(outcome=rownames(b[[29]]), b[[29]])
df.b30<-data.frame(outcome=rownames(b[[30]]), b[[30]])
df.b31<-data.frame(outcome=rownames(b[[31]]), b[[31]])
```

Still in loop: Bind df.aa to the df.bs. So now, the df.cs have the relevant variables for interpreting/labeling results (foodgroup, age/sex/race, etc…, all the variables listed in vars.to.keep.for.df.a)

```
df.c1<-cbind(df.aa, df.b1)
df.c2<-cbind(df.aa, df.b2)
df.c3<-cbind(df.aa, df.b3)
df.c4<-cbind(df.aa, df.b4)
df.c5<-cbind(df.aa, df.b5)

df.c6<-cbind(df.aa, df.b6)
df.c7<-cbind(df.aa, df.b7)
df.c8<-cbind(df.aa, df.b8)
df.c9<-cbind(df.aa, df.b9)

df.c10<-cbind(df.aa, df.b10)
df.c11<-cbind(df.aa, df.b11)
df.c12<-cbind(df.aa, df.b12)
df.c13<-cbind(df.aa, df.b13)

df.c14<-cbind(df.aa, df.b14)
df.c15<-cbind(df.aa, df.b15)
df.c16<-cbind(df.aa, df.b16)
df.c17<-cbind(df.aa, df.b17)
df.c18<-cbind(df.aa, df.b18)
df.c19<-cbind(df.aa, df.b19)

df.c20<-cbind(df.aa, df.b20)
df.c21<-cbind(df.aa, df.b21)
df.c22<-cbind(df.aa, df.b22)
df.c23<-cbind(df.aa, df.b23)

df.c24<-cbind(df.aa, df.b24)
df.c25<-cbind(df.aa, df.b25)
df.c26<-cbind(df.aa, df.b26)
df.c27<-cbind(df.aa, df.b27)

df.c28<-cbind(df.aa, df.b28)
df.c29<-cbind(df.aa, df.b29)
df.c30<-cbind(df.aa, df.b30)
df.c31<-cbind(df.aa, df.b31)
```

Still in loop: Create “impact_unit_strings” string vector, which is just the “outcome” variable with “_unit” appended to it. Then, create “outcome_unit” variable for all df.cs. Then, for each df.c, loop through each row and populate “outcome_unit” with the the corresponding unit for that outcome for that row (which is extracted from the row j and column impact_unit_strings[j] for the df.c).

```
impact_unit_strings<-paste(df.b1$outcome, "_unit", sep="")

df.c1$outcome_unit<-NA
df.c2$outcome_unit<-NA
df.c3$outcome_unit<-NA
df.c4$outcome_unit<-NA
df.c5$outcome_unit<-NA

df.c6$outcome_unit<-NA
df.c7$outcome_unit<-NA
df.c8$outcome_unit<-NA
df.c9$outcome_unit<-NA

df.c10$outcome_unit<-NA
df.c11$outcome_unit<-NA
df.c12$outcome_unit<-NA
df.c13$outcome_unit<-NA

df.c14$outcome_unit<-NA
df.c15$outcome_unit<-NA
df.c16$outcome_unit<-NA
df.c17$outcome_unit<-NA
df.c18$outcome_unit<-NA
df.c19$outcome_unit<-NA

df.c20$outcome_unit<-NA
df.c21$outcome_unit<-NA
df.c22$outcome_unit<-NA
df.c23$outcome_unit<-NA

df.c24$outcome_unit<-NA
df.c25$outcome_unit<-NA
df.c26$outcome_unit<-NA
df.c27$outcome_unit<-NA

df.c28$outcome_unit<-NA
df.c29$outcome_unit<-NA
df.c30$outcome_unit<-NA
df.c31$outcome_unit<-NA

for(j in 1:dim(df.c1)[1])
{
  df.c1$outcome_unit[j]<-df.c1[j,impact_unit_strings[j]]
  df.c2$outcome_unit[j]<-df.c2[j,impact_unit_strings[j]]
  df.c3$outcome_unit[j]<-df.c3[j,impact_unit_strings[j]]
  df.c4$outcome_unit[j]<-df.c4[j,impact_unit_strings[j]]
  df.c5$outcome_unit[j]<-df.c5[j,impact_unit_strings[j]]

  df.c6$outcome_unit[j]<-df.c6[j,impact_unit_strings[j]]
  df.c7$outcome_unit[j]<-df.c7[j,impact_unit_strings[j]]
  df.c8$outcome_unit[j]<-df.c8[j,impact_unit_strings[j]]
  df.c9$outcome_unit[j]<-df.c9[j,impact_unit_strings[j]]

  df.c10$outcome_unit[j]<-df.c10[j,impact_unit_strings[j]]
```

```
    df.c11$outcome_unit[j]<-df.c11[j,impact_unit_strings[j]]
    df.c12$outcome_unit[j]<-df.c12[j,impact_unit_strings[j]]
    df.c13$outcome_unit[j]<-df.c13[j,impact_unit_strings[j]]

    df.c14$outcome_unit[j]<-df.c14[j,impact_unit_strings[j]]
    df.c15$outcome_unit[j]<-df.c15[j,impact_unit_strings[j]]
    df.c16$outcome_unit[j]<-df.c16[j,impact_unit_strings[j]]
    df.c17$outcome_unit[j]<-df.c17[j,impact_unit_strings[j]]
    df.c18$outcome_unit[j]<-df.c18[j,impact_unit_strings[j]]
    df.c19$outcome_unit[j]<-df.c19[j,impact_unit_strings[j]]

    df.c20$outcome_unit[j]<-df.c20[j,impact_unit_strings[j]]
    df.c21$outcome_unit[j]<-df.c21[j,impact_unit_strings[j]]
    df.c22$outcome_unit[j]<-df.c22[j,impact_unit_strings[j]]
    df.c23$outcome_unit[j]<-df.c23[j,impact_unit_strings[j]]

    df.c24$outcome_unit[j]<-df.c24[j,impact_unit_strings[j]]
    df.c25$outcome_unit[j]<-df.c25[j,impact_unit_strings[j]]
    df.c26$outcome_unit[j]<-df.c26[j,impact_unit_strings[j]]
    df.c27$outcome_unit[j]<-df.c27[j,impact_unit_strings[j]]

    df.c28$outcome_unit[j]<-df.c28[j,impact_unit_strings[j]]
    df.c29$outcome_unit[j]<-df.c29[j,impact_unit_strings[j]]
    df.c30$outcome_unit[j]<-df.c30[j,impact_unit_strings[j]]
    df.c31$outcome_unit[j]<-df.c31[j,impact_unit_strings[j]]
  }
```

Still in loop: move the new “outcome_unit” column to right after “outcome”, and get rid of the impact factor specific columns for units.

```
df.c1<- df.c1 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings) ##move the
df.c2<- df.c2 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c3<- df.c3 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c4<- df.c4 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c5<- df.c5 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)

df.c6<- df.c6 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c7<- df.c7 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c8<- df.c8 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c9<- df.c9 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)

df.c10<- df.c10 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c11<- df.c11 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c12<- df.c12 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c13<- df.c13 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)

df.c14<- df.c14 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c15<- df.c15 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c16<- df.c16 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c17<- df.c17 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c18<- df.c18 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c19<- df.c19 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)

df.c20<- df.c20 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c21<- df.c21 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c22<- df.c22 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c23<- df.c23 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)

df.c24<- df.c24 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c25<- df.c25 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c26<- df.c26 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c27<- df.c27 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)

df.c28<- df.c28 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c29<- df.c29 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c30<- df.c30 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
df.c31<- df.c31 %>% relocate(outcome_unit, .after=outcome) %>% select(-impact_unit_strings)
```

We assign each data frame we made to be the i-th element of their corresponding lists. Recall, earlier in the code, we made 31 different lists to store output (one for each kind of output type). We are now populating that list, the i-th element corresponding to the appropriate df.c. (e.g the current version of df.c1 for this iteration within the for loop is being stored as the i-th element of list.of.impact.sims.total, and so on).

```
  list.of.impact.sims.total[[i]]<-df.c1
  list.of.substitution.impact.sims.total[[i]]<-df.c2
  list.of.combined.impact.sims.total[[i]]<-df.c3
  list.of.population.sims[[i]]<-df.c4
  list.of.delta.sims.total[[i]]<-df.c5

  list.of.impact.sims.consumed[[i]]<-df.c6
  list.of.substitution.impact.sims.consumed[[i]]<-df.c7
  list.of.combined.impact.sims.consumed[[i]]<-df.c8
  list.of.delta.sims.consumed[[i]]<-df.c9

  list.of.impact.sims.unconsumed[[i]]<-df.c10
  list.of.substitution.impact.sims.unconsumed[[i]]<-df.c11
  list.of.combined.impact.sims.unconsumed[[i]]<-df.c12
  list.of.delta.sims.unconsumed[[i]]<-df.c13

  list.of.current.intake.impact.sims.total[[i]]<-df.c14
  list.of.CF.intake.impact.sims.total[[i]]<-df.c15
  list.of.current.intake.impact.sims.consumed[[i]]<-df.c16
  list.of.CF.intake.impact.sims.consumed[[i]]<-df.c17
  list.of.current.intake.impact.sims.unconsumed[[i]]<-df.c18
  list.of.CF.intake.impact.sims.unconsumed[[i]]<-df.c19

  list.of.impact.sims.inedible[[i]]<-df.c20
  list.of.substitution.impact.sims.inedible[[i]]<-df.c21
  list.of.combined.impact.sims.inedible[[i]]<-df.c22
  list.of.delta.sims.inedible[[i]]<-df.c23

  list.of.impact.sims.wasted[[i]]<-df.c24
  list.of.substitution.impact.sims.wasted[[i]]<-df.c25
  list.of.combined.impact.sims.wasted[[i]]<-df.c26
  list.of.delta.sims.wasted[[i]]<-df.c27

  list.of.current.intake.impact.sims.inedible[[i]]<-df.c28
  list.of.CF.intake.impact.sims.inedible[[i]]<-df.c29
  list.of.current.intake.impact.sims.wasted[[i]]<-df.c30
  list.of.CF.intake.impact.sims.wasted[[i]]<-df.c31
```

End the loop.

```
}
```

For each list, row-bind the elements of the list (corresponding to a subgroup and dietary factor) together into a data frame. Each of these data frames has results for all dietary factors and subgroups, for one outcome (specified in name of data frame). NAs are also replaced with 0.

Recall there are 31 lists,each corresponding to some combination of 1. what we are looking at (impact (as in, impact of change), substitution impact, combined impact, population, delta (aka, change in intake), current intake impact, counterfactual intake impact), and 2. level of disaggregation (total, consumed, unconsumed, inedible, wasted).

```
##not sure why I went with full joint before, rbind is much faster, and shouldn't cause problems.

# added NaN removal on 10/6
impact.sims.total.allgroups<-Reduce(rbind,list.of.impact.sims.total) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
substitution.impact.sims.total.allgroups<-Reduce(rbind,list.of.substitution.impact.sims.total) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
combined.impact.sims.total.allgroups<-Reduce(rbind,list.of.combined.impact.sims.total) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
population.sims.allgroups<-Reduce(rbind,list.of.population.sims) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
delta.sims.total.allgroups<-Reduce(rbind,list.of.delta.sims.total) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))

impact.sims.consumed.allgroups<-Reduce(rbind,list.of.impact.sims.consumed) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
substitution.impact.sims.consumed.allgroups<-Reduce(rbind,list.of.substitution.impact.sims.consumed) %>
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
combined.impact.sims.consumed.allgroups<-Reduce(rbind,list.of.combined.impact.sims.consumed) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
delta.sims.consumed.allgroups<-Reduce(rbind,list.of.delta.sims.consumed) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))

impact.sims.unconsumed.allgroups<-Reduce(rbind,list.of.impact.sims.unconsumed) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
substitution.impact.sims.unconsumed.allgroups<-Reduce(rbind,list.of.substitution.impact.sims.unconsumed
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
combined.impact.sims.unconsumed.allgroups<-Reduce(rbind,list.of.combined.impact.sims.unconsumed) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
delta.sims.unconsumed.allgroups<-Reduce(rbind,list.of.delta.sims.unconsumed) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))

current.intake.impact.sims.total.allgroups<-Reduce(rbind,list.of.current.intake.impact.sims.total) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
CF.intake.impact.sims.total.allgroups<-Reduce(rbind,list.of.CF.intake.impact.sims.total) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
current.intake.impact.sims.consumed.allgroups<-Reduce(rbind,list.of.current.intake.impact.sims.consumed
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
CF.intake.impact.sims.consumed.allgroups<-Reduce(rbind,list.of.CF.intake.impact.sims.consumed) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
current.intake.impact.sims.unconsumed.allgroups<-Reduce(rbind,list.of.current.intake.impact.sims.uncons
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
CF.intake.impact.sims.unconsumed.allgroups<-Reduce(rbind,list.of.CF.intake.impact.sims.unconsumed) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))

impact.sims.inedible.allgroups<-Reduce(rbind,list.of.impact.sims.inedible) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
substitution.impact.sims.inedible.allgroups<-Reduce(rbind,list.of.substitution.impact.sims.inedible) %>
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
combined.impact.sims.inedible.allgroups<-Reduce(rbind,list.of.combined.impact.sims.inedible) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
delta.sims.inedible.allgroups<-Reduce(rbind,list.of.delta.sims.inedible) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
```

```
impact.sims.wasted.allgroups<-Reduce(rbind,list.of.impact.sims.wasted) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
substitution.impact.sims.wasted.allgroups<-Reduce(rbind,list.of.substitution.impact.sims.wasted) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
combined.impact.sims.wasted.allgroups<-Reduce(rbind,list.of.combined.impact.sims.wasted) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
delta.sims.wasted.allgroups<-Reduce(rbind,list.of.delta.sims.wasted) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))

current.intake.impact.sims.inedible.allgroups<-Reduce(rbind,list.of.current.intake.impact.sims.inedible
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
CF.intake.impact.sims.inedible.allgroups<-Reduce(rbind,list.of.CF.intake.impact.sims.inedible) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
current.intake.impact.sims.wasted.allgroups<-Reduce(rbind,list.of.current.intake.impact.sims.wasted) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
CF.intake.impact.sims.wasted.allgroups<-Reduce(rbind,list.of.CF.intake.impact.sims.wasted) %>%
  mutate(across(everything(), ~replace(.x, is.nan(.x), 0)))
```

Save dataframes into csv files.

```
#setwd("C:\\Users\\fcudhe01\\Box\\lasting_aim_3\\data\\out\\example_cost\\per_capita")
setwd(output_location)
fwrite(x=impact.sims.percapita.total.allgroups, file="per_capita/costenv.sims.percapita.granular.csv")
fwrite(x=substitution.impact.sims.percapita.total.allgroups, file="per_capita/substitution.costenv.sims
fwrite(x=combined.impact.percapita.total.allgroups, file="per_capita/combined.costenv.sims.percapita.gr
fwrite(x=delta.percapita.allgroups, file="per_capita/delta.costenv.sims.percapita.granular.csv")

fwrite(x=impact.sims.percapita.consumed.allgroups, file="per_capita/costenv.sims.percapita.consumed.gra
fwrite(x=substitution.impact.sims.percapita.consumed.allgroups, file="per_capita/substitution.costenv.s
fwrite(x=combined.impact.percapita.consumed.allgroups, file="per_capita/combined.costenv.sims.percapita

fwrite(x=impact.sims.percapita.unconsumed.allgroups, file="per_capita/costenv.sims.percapita.unconsumed
fwrite(x=substitution.impact.sims.percapita.unconsumed.allgroups, file="per_capita/substitution.costenv
fwrite(x=combined.impact.percapita.unconsumed.allgroups, file="per_capita/combined.costenv.sims.percapi

fwrite(x=current.intake.impact.sims.percapita.total.allgroups, file="per_capita/current.intake.costenv.
fwrite(x=CF.intake.impact.sims.percapita.total.allgroups, file="per_capita/CF.costenv.intake.sims.perca
fwrite(x=current.intake.impact.sims.percapita.consumed.allgroups, file="per_capita/current.intake.coste
fwrite(x=CF.intake.impact.sims.percapita.consumed.allgroups, file="per_capita/CF.intake.costenv.sims.pe
fwrite(x=current.intake.impact.sims.percapita.unconsumed.allgroups, file="per_capita/current.intake.cos
fwrite(x=CF.intake.impact.sims.percapita.unconsumed.allgroups, file="per_capita/CF.intake.costenv.sims.

fwrite(x=impact.sims.percapita.consumed.allgroups, file="per_capita/costenv.sims.percapita.inedible.gra
fwrite(x=substitution.impact.sims.percapita.consumed.allgroups, file="per_capita/substitution.costenv.s
fwrite(x=combined.impact.percapita.consumed.allgroups, file="per_capita/combined.costenv.sims.percapita

fwrite(x=impact.sims.percapita.consumed.allgroups, file="per_capita/costenv.sims.percapita.wasted.granu
fwrite(x=substitution.impact.sims.percapita.consumed.allgroups, file="per_capita/substitution.costenv.s
fwrite(x=combined.impact.percapita.consumed.allgroups, file="per_capita/combined.costenv.sims.percapita

fwrite(x=current.intake.impact.sims.percapita.consumed.allgroups, file="per_capita/current.intake.coste
fwrite(x=CF.intake.impact.sims.percapita.consumed.allgroups, file="per_capita/CF.intake.costenv.sims.pe
fwrite(x=current.intake.impact.sims.percapita.unconsumed.allgroups, file="per_capita/current.intake.cos
fwrite(x=CF.intake.impact.sims.percapita.unconsumed.allgroups, file="per_capita/CF.intake.costenv.sims.
```

All of what we just did pertains to population level impact. Next. we get per capita sims for all outputs of interest.

Recall there are 31 outcome types of interest, each corresponding to some combination of 1. what we are looking at (impact (as in, impact of change), substitution impact, combined impact, population, delta (aka, change in intake), current intake impact, counterfactual intake impact), and 2. level of disaggregation (total, consumed, unconsumed, inedible, wasted).

```
### per capita
impact.sims.percapita.total.allgroups<-get.percapita.sims(impact.sims=impact.sims.total.allgroups,
                                                          population.sims=population.sims.allgroups,
                                                          pop.sims.names=paste("X", c(1:n.sims), sep=""),
                                                          new.pop.sims.names=paste("pop", c(1:n.sims), sep=""
                                                          sims.names=paste("X", c(1:n.sims), sep=""))
substitution.impact.sims.percapita.total.allgroups<-get.percapita.sims(impact.sims=substitution.impact.
                                                                       population.sims=population.sims.allgro
                                                                       pop.sims.names=paste("X", c(1:n.sims),
                                                                       new.pop.sims.names=paste("pop", c(1:n.
                                                                       sims.names=paste("X", c(1:n.sims), sep
combined.impact.percapita.total.allgroups<-get.percapita.sims(impact.sims=combined.impact.sims.total.al
                                                              population.sims=population.sims.allgroups,
                                                              pop.sims.names=paste("X", c(1:n.sims), sep=""),
                                                              new.pop.sims.names=paste("pop", c(1:n.sims), se
                                                              sims.names=paste("X", c(1:n.sims), sep=""))
delta.percapita.allgroups<-get.percapita.sims(impact.sims=delta.sims.total.allgroups,
                                              population.sims=population.sims.allgroups,
                                              pop.sims.names=paste("X", c(1:n.sims), sep=""),
                                              new.pop.sims.names=paste("pop", c(1:n.sims), sep=""),
                                              sims.names=paste("X", c(1:n.sims), sep=""))

impact.sims.percapita.consumed.allgroups<-get.percapita.sims(impact.sims=impact.sims.consumed.allgroups
                                                          population.sims=population.sims.allgroups,
                                                          pop.sims.names=paste("X", c(1:n.sims), sep=""),
                                                          new.pop.sims.names=paste("pop", c(1:n.sims), sep=""
                                                          sims.names=paste("X", c(1:n.sims), sep=""))
substitution.impact.sims.percapita.consumed.allgroups<-get.percapita.sims(impact.sims=substitution.impa
                                                                       population.sims=population.sims.allgro
                                                                       pop.sims.names=paste("X", c(1:n.sims),
                                                                       new.pop.sims.names=paste("pop", c(1:n.
                                                                       sims.names=paste("X", c(1:n.sims), sep
combined.impact.percapita.consumed.allgroups<-get.percapita.sims(impact.sims=combined.impact.sims.consu
                                                              population.sims=population.sims.allgroups,
                                                              pop.sims.names=paste("X", c(1:n.sims), sep=""),
                                                              new.pop.sims.names=paste("pop", c(1:n.sims), se
                                                              sims.names=paste("X", c(1:n.sims), sep=""))

impact.sims.percapita.unconsumed.allgroups<-get.percapita.sims(impact.sims=impact.sims.unconsumed.allgr
                                                                   population.sims=population.sims.allgroups,
                                                                   pop.sims.names=paste("X", c(1:n.sims), sep
                                                                   new.pop.sims.names=paste("pop", c(1:n.sims
                                                                   sims.names=paste("X", c(1:n.sims), sep="")
substitution.impact.sims.percapita.unconsumed.allgroups<-get.percapita.sims(impact.sims=substitution.im
                                                                                population.sims=population.si
                                                                                pop.sims.names=paste("X", c(1
                                                                                new.pop.sims.names=paste("pop
                                                                                sims.names=paste("X", c(1:n.s
combined.impact.percapita.unconsumed.allgroups<-get.percapita.sims(impact.sims=combined.impact.sims.unc
                                                                       population.sims=population.sims.allgro
                                                                       pop.sims.names=paste("X", c(1:n.sims),
                                                                       new.pop.sims.names=paste("pop", c(1:n.
                                                                       sims.names=paste("X", c(1:n.sims), sep
```

```
current.intake.impact.sims.percapita.total.allgroups<-get.percapita.sims(impact.sims=current.intake.imp
                                                          population.sims=population.sims.allgroups,
                                                          pop.sims.names=paste("X", c(1:n.sims), sep=""
                                                          new.pop.sims.names=paste("pop", c(1:n.sims),
                                                          sims.names=paste("X", c(1:n.sims), sep=""))
CF.intake.impact.sims.percapita.total.allgroups<-get.percapita.sims(impact.sims=CF.intake.impact.sims.t
                                                             population.sims=population.sims.allgroups,
                                                             pop.sims.names=paste("X", c(1:n.sims), sep
                                                             new.pop.sims.names=paste("pop", c(1:n.sims
                                                             sims.names=paste("X", c(1:n.sims), sep="")
current.intake.impact.sims.percapita.consumed.allgroups<-get.percapita.sims(impact.sims=current.intake.
                                                                         population.sims=population.sim
                                                                         pop.sims.names=paste("X", c(1:
                                                                         new.pop.sims.names=paste("pop"
                                                                         sims.names=paste("X", c(1:n.si
CF.intake.impact.sims.percapita.consumed.allgroups<-get.percapita.sims(impact.sims=CF.intake.impact.sim
                                                                    population.sims=population.sims.all
                                                                    pop.sims.names=paste("X", c(1:n.sim
                                                                    new.pop.sims.names=paste("pop", c(1
                                                                    sims.names=paste("X", c(1:n.sims),
current.intake.impact.sims.percapita.unconsumed.allgroups<-get.percapita.sims(impact.sims=current.intak
                                                                            population.sims=population.
                                                                            pop.sims.names=paste("X", c
                                                                            new.pop.sims.names=paste("p
                                                                            sims.names=paste("X", c(1:n
CF.intake.impact.sims.percapita.unconsumed.allgroups<-get.percapita.sims(impact.sims=CF.intake.impact.s
                                                                       population.sims=population.sims.
                                                                       pop.sims.names=paste("X", c(1:n.
                                                                       new.pop.sims.names=paste("pop",
                                                                       sims.names=paste("X", c(1:n.sims


impact.sims.percapita.inedible.allgroups<-get.percapita.sims(impact.sims=impact.sims.inedible.allgroups
                                                               population.sims=population.sims.allgroup
                                                               pop.sims.names=paste("X", c(1:n.sims), s
                                                               new.pop.sims.names=paste("pop", c(1:n.si
                                                               sims.names=paste("X", c(1:n.sims), sep="
substitution.impact.sims.percapita.inedible.allgroups<-get.percapita.sims(impact.sims=substitution.impa
                                                                            population.sims=population.
                                                                            pop.sims.names=paste("X", c
                                                                            new.pop.sims.names=paste("p
                                                                            sims.names=paste("X", c(1:n
combined.impact.percapita.inedible.allgroups<-get.percapita.sims(impact.sims=combined.impact.sims.inedi
                                                                   population.sims=population.sims.allg
                                                                   pop.sims.names=paste("X", c(1:n.sims
                                                                   new.pop.sims.names=paste("pop", c(1:
                                                                   sims.names=paste("X", c(1:n.sims), s

impact.sims.percapita.wasted.allgroups<-get.percapita.sims(impact.sims=impact.sims.wasted.allgroups,
                                                             population.sims=population.sims.allgroups,
                                                             pop.sims.names=paste("X", c(1:n.sims), sep
                                                             new.pop.sims.names=paste("pop", c(1:n.sims
                                                             sims.names=paste("X", c(1:n.sims), sep="")
substitution.impact.sims.percapita.wasted.allgroups<-get.percapita.sims(impact.sims=substitution.impact
```

```
                                                                           population.sims=population.si
                                                                           pop.sims.names=paste("X", c(1
                                                                           new.pop.sims.names=paste("pop
                                                                           sims.names=paste("X", c(1:n.s
combined.impact.percapita.wasted.allgroups<-get.percapita.sims(impact.sims=combined.impact.sims.wasted.
                                                                  population.sims=population.sims.allgro
                                                                  pop.sims.names=paste("X", c(1:n.sims),
                                                                  new.pop.sims.names=paste("pop", c(1:n.
                                                                  sims.names=paste("X", c(1:n.sims), sep

current.intake.impact.sims.percapita.inedible.allgroups<-get.percapita.sims(impact.sims=current.intake.
                                                                             population.sims=population.
                                                                             pop.sims.names=paste("X", c
                                                                             new.pop.sims.names=paste("p
                                                                             sims.names=paste("X", c(1:n
CF.intake.impact.sims.percapita.inedible.allgroups<-get.percapita.sims(impact.sims=CF.intake.impact.sim
                                                                        population.sims=population.sims.
                                                                        pop.sims.names=paste("X", c(1:n.
                                                                        new.pop.sims.names=paste("pop",
                                                                        sims.names=paste("X", c(1:n.sims
current.intake.impact.sims.percapita.wasted.allgroups<-get.percapita.sims(impact.sims=current.intake.im
                                                                               population.sims=populatio
                                                                               pop.sims.names=paste("X",
                                                                               new.pop.sims.names=paste(
                                                                               sims.names=paste("X", c(1
CF.intake.impact.sims.percapita.wasted.allgroups<-get.percapita.sims(impact.sims=CF.intake.impact.sims.
                                                                          population.sims=population.sim
                                                                          pop.sims.names=paste("X", c(1:
                                                                          new.pop.sims.names=paste("pop"
                                                                          sims.names=paste("X", c(1:n.si
```

Save them into csv files.

```
#setwd("C:\\Users\\fcudhe01\\Box\\lasting_aim_3\\data\\out\\example_cost\\per_capita")
setwd(output_location)
fwrite(x=impact.sims.percapita.total.allgroups, file="per_capita/costenv.sims.percapita.granular.csv")
fwrite(x=substitution.impact.sims.percapita.total.allgroups, file="per_capita/substitution.costenv.sims
fwrite(x=combined.impact.percapita.total.allgroups, file="per_capita/combined.costenv.sims.percapita.gr
fwrite(x=delta.percapita.allgroups, file="per_capita/delta.costenv.sims.percapita.granular.csv")

fwrite(x=impact.sims.percapita.consumed.allgroups, file="per_capita/costenv.sims.percapita.consumed.gra
fwrite(x=substitution.impact.sims.percapita.consumed.allgroups, file="per_capita/substitution.costenv.s
fwrite(x=combined.impact.percapita.consumed.allgroups, file="per_capita/combined.costenv.sims.percapita

fwrite(x=impact.sims.percapita.unconsumed.allgroups, file="per_capita/costenv.sims.percapita.unconsumed
fwrite(x=substitution.impact.sims.percapita.unconsumed.allgroups, file="per_capita/substitution.costenv
fwrite(x=combined.impact.percapita.unconsumed.allgroups, file="per_capita/combined.costenv.sims.percapi

fwrite(x=current.intake.impact.sims.percapita.total.allgroups, file="per_capita/current.intake.costenv.
fwrite(x=CF.intake.impact.sims.percapita.total.allgroups, file="per_capita/CF.costenv.intake.sims.perca
fwrite(x=current.intake.impact.sims.percapita.consumed.allgroups, file="per_capita/current.intake.coste
fwrite(x=CF.intake.impact.sims.percapita.consumed.allgroups, file="per_capita/CF.intake.costenv.sims.pe
fwrite(x=current.intake.impact.sims.percapita.unconsumed.allgroups, file="per_capita/current.intake.cos
fwrite(x=CF.intake.impact.sims.percapita.unconsumed.allgroups, file="per_capita/CF.intake.costenv.sims.

fwrite(x=impact.sims.percapita.consumed.allgroups, file="per_capita/costenv.sims.percapita.inedible.gra
fwrite(x=substitution.impact.sims.percapita.consumed.allgroups, file="per_capita/substitution.costenv.s
fwrite(x=combined.impact.percapita.consumed.allgroups, file="per_capita/combined.costenv.sims.percapita

fwrite(x=impact.sims.percapita.consumed.allgroups, file="per_capita/costenv.sims.percapita.wasted.granu
fwrite(x=substitution.impact.sims.percapita.consumed.allgroups, file="per_capita/substitution.costenv.s
fwrite(x=combined.impact.percapita.consumed.allgroups, file="per_capita/combined.costenv.sims.percapita

fwrite(x=current.intake.impact.sims.percapita.consumed.allgroups, file="per_capita/current.intake.coste
fwrite(x=CF.intake.impact.sims.percapita.consumed.allgroups, file="per_capita/CF.intake.costenv.sims.pe
fwrite(x=current.intake.impact.sims.percapita.unconsumed.allgroups, file="per_capita/current.intake.cos
fwrite(x=CF.intake.impact.sims.percapita.unconsumed.allgroups, file="per_capita/CF.intake.costenv.sims.
```

Get summary stats from simulations (mean, median, standard deviation, uncertainty intervals, the things you'll be reporting in your manuscript) for all outcome of interests and save (for both total and per capita impact).

```
################
##get summary stats
#############
all.impact.sims.summary<-get.summary.stats(impact.sims=impact.sims.total.allgroups,
                                           substitution.impact.sims=substitution.impact.sims.total.allg
                                           combined.impact.sims=combined.impact.sims.total.allgroups,
                                           delta.sims=delta.sims.total.allgroups,
                                           impact.sims.consumed=impact.sims.consumed.allgroups,
                                           substitution.impact.sims.consumed=substitution.impact.sims.c
                                           combined.impact.sims.consumed=combined.impact.sims.consumed.
                                           impact.sims.unconsumed=impact.sims.unconsumed.allgroups,
                                           substitution.impact.sims.unconsumed=substitution.impact.sims
                                           combined.impact.sims.unconsumed=combined.impact.sims.unconsu
                                           current.intake.impact.sims=current.intake.impact.sims.total.
                                           current.intake.impact.sims.consumed=current.intake.impact.si
                                           current.intake.impact.sims.unconsumed=current.intake.impact.
                                           CF.intake.impact.sims=CF.intake.impact.sims.total.allgroups,
                                           CF.intake.impact.sims.consumed=CF.intake.impact.sims.consume
                                           CF.intake.impact.sims.unconsumed=CF.intake.impact.sims.uncon
                                           impact.sims.inedible=impact.sims.inedible.allgroups,
                                           substitution.impact.sims.inedible=substitution.impact.sims.i
                                           combined.impact.sims.inedible=combined.impact.sims.inedible.
                                           impact.sims.wasted=impact.sims.wasted.allgroups,
                                           substitution.impact.sims.wasted=substitution.impact.sims.was
                                           combined.impact.sims.wasted=combined.impact.sims.wasted.allg
                                           current.intake.impact.sims.inedible=current.intake.impact.si
                                           current.intake.impact.sims.wasted=current.intake.impact.sims
                                           CF.intake.impact.sims.inedible=CF.intake.impact.sims.inedibl
                                           CF.intake.impact.sims.wasted=CF.intake.impact.sims.wasted.al
                                           vars=paste("X", 1:n.sims, sep=""))
#setwd("C:\\Users\\fcudhe01\\Box\\lasting_aim_3\\data\\out\\example_cost")
setwd(output_location)
write.csv(x=all.impact.sims.summary, file="costenv.summary.granular.csv")

##
#now use the function for per capita results
all.impact.sims.percapita.summary<-get.summary.stats(impact.sims=impact.sims.percapita.total.allgroups,
                                                     substitution.impact.sims=substitution.impact.sims.
                                                     combined.impact.sims=combined.impact.percapita.tot
                                                     delta.sims=delta.percapita.allgroups,
                                                     impact.sims.consumed=impact.sims.percapita.consume
                                                     substitution.impact.sims.consumed=substitution.imp
                                                     combined.impact.sims.consumed=combined.impact.perc
                                                     impact.sims.unconsumed=impact.sims.percapita.uncon
                                                     substitution.impact.sims.unconsumed=substitution.i
                                                     combined.impact.sims.unconsumed=combined.impact.pe
                                                     current.intake.impact.sims=current.intake.impact.s
                                                     current.intake.impact.sims.consumed=current.intake
                                                     current.intake.impact.sims.unconsumed=current.inta
                                                     CF.intake.impact.sims=CF.intake.impact.sims.total.
                                                     CF.intake.impact.sims.consumed=CF.intake.impact.si
                                                     CF.intake.impact.sims.unconsumed=CF.intake.impact.
                                                     impact.sims.inedible=impact.sims.percapita.inedibl
                                                     substitution.impact.sims.inedible=substitution.imp
```

```
                                                    combined.impact.sims.inedible=combined.impact.perc
                                                    impact.sims.wasted=impact.sims.percapita.wasted.al
                                                    substitution.impact.sims.wasted=substitution.impac
                                                    combined.impact.sims.wasted=combined.impact.percap
                                                    current.intake.impact.sims.inedible=current.intake
                                                    current.intake.impact.sims.wasted=current.intake.i
                                                    CF.intake.impact.sims.inedible=CF.intake.impact.si
                                                    CF.intake.impact.sims.wasted=CF.intake.impact.sims
                                                    vars=paste("X", 1:n.sims, sep=""))
#setwd("C:\\Users\\fcudhe01\\Box\\lasting_aim_3\\data\\out\\example_cost\\per_capita")
setwd(output_location)
write.csv(x=all.impact.sims.percapita.summary, file="per_capita/costenv.percapita.summary.granular.csv"
```

Next, we want to get summary stat for all possible strata combinations (not just the most granular, age/sex/race/foodgroup).

First we create a list of all possible strata combinations. Note, the string vector “strata” contains all the strata being used for the analysis. If you want to use different strata, you need to change the line defining “strata” accordingly.

```
#########get sums based on less granular subgroups#########
##get all possible combinations of strata
strata<-c("age_gp", "sex_gp", "race_gp", "Foodgroup")

strata.combos<-list()
for(i in 1:length(strata))
{
  x<-combn(strata, i)
  combos.for.i<-split(x, rep(1:ncol(x), each = nrow(x))) #https://stackoverflow.com/questions/6819804/h
  strata.combos<-c(strata.combos, combos.for.i)
}
strata.combos["null"]<-list(NULL) ##add null element to list to use for overall numbers
```

Get summary stats by all strata combinations for all outcomes of interest. Note that the objects being created here are lists of lists. The second level are the different strata combinations. The first level denotes outcome type: 1 = total summary stats, 2 = total sims, 3 = per capita summary stats, 4 = per capita sims. That means, for example, that summ.stats.by.strata.cost is a list, and each element within that list is also a list. Specifically, it is a list of 4 lists (each one denoting outcome type), and each of those 4 lists is itself a list of 48 data frames, one for each subgroup.

```
# summary.stats.by.strata<-function(impact.sims.allgroups, strata.combos.list, pop.sims)
# {
#   cols.to.sum<-paste("X", 1:n.sims, sep="")
#   pop.cols.to.sum<-paste("pop", 1:n.sims, sep="")
#
#   sims.data.table<-as.data.table(impact.sims.allgroups)
#   pop.sims.data.table<-as.data.table(pop.sims)
#
#   strata.combos.sims<-list()
#   strata.combos.sims.long<-list()
#   strata.combos.summary.stats.long<-list()
#   strata.combos.summary.stats.wide<-list()
#
#   pop.strata.combos.sims<-list()
#
#   percapita.sims<-list()
#   percapita.sims.long<-list()
#   percapita.summary.stats.long<-list()
#   percapita.summary.stats.wide<-list()
#
#   for(j in 1:length(strata.combos))
#   {
#     labels<-c(strata.combos[[j]], "intake_units", "outcome", "outcome_unit")
#
#     strata.combos.sims[[j]]<-sims.data.table[,lapply(.SD, sum), by=c(strata.combos[[j]], "intake_unit
#                                            .SDcols=cols.to.sum]
#
#     pop.strata<-strata.combos[[j]][!(strata.combos[[j]] %in% c("Foodgroup", "broad_foodgroup"))]
#
#     if(is.null(pop.strata))
#     {
#       pop.strata.combos.sims[[j]]<-pop.sims.data.table[,lapply(.SD, sum), .SDcols=cols.to.sum]
#     }else
#     {
#       pop.strata.combos.sims[[j]]<-pop.sims.data.table[,lapply(.SD, sum), by=c(pop.strata), .SDcols=c
#     }
#     names(pop.strata.combos.sims[[j]])<-gsub(x=names(pop.strata.combos.sims[[j]]), pattern="X", repla
#
#     if(length(intersect(names(strata.combos.sims[[j]]), names(pop.strata.combos.sims[[j]])))==0)
#     {
#       merged<-cbind(strata.combos.sims[[j]], pop.strata.combos.sims[[j]])
#     }
#     if(length(intersect(names(strata.combos.sims[[j]]), names(pop.strata.combos.sims[[j]])))!=0)
#     {
#       merged<-merge(strata.combos.sims[[j]], pop.strata.combos.sims[[j]])
#     }
#
#
#     merged.names<-merged[,..labels]
#     percapita.sims[[j]]<-merged[,..cols.to.sum]/merged[,..pop.cols.to.sum]
#     percapita.sims[[j]]<-cbind(merged.names, percapita.sims[[j]])
#
#     strata.combos.sims.long[[j]]<-melt(strata.combos.sims[[j]], id.vars = c(strata.combos[[j]], "inta
#     strata.combos.summary.stats.long[[j]]<-strata.combos.sims.long[[j]][, .(quantile(value, c(.025, .
```

```
#     strata.combos.summary.stats.long[[j]]<-cbind(strata.combos.summary.stats.long[[j]],
#                                              as.factor(rep(c("lower_bound (2.5th percentile)", "m
#     #strata.combos.summary.stats.wide[[j]]<-dcast(strata.combos.summary.stats.long[[j]], "Comorbid_co
#     strata.combos.summary.stats.wide[[j]]<-dcast(strata.combos.summary.stats.long[[j]], paste(paste(s
#                                              value.var="V1")
#
#     ##nperhaps in future, make it so that the function doesn't rely on specific names for intake_unit
#
#     percapita.sims.long[[j]]<-melt(percapita.sims[[j]], id.vars = c(strata.combos[[j]], "intake_units
#     percapita.summary.stats.long[[j]]<-percapita.sims.long[[j]][, .(quantile(value, c(.025, .5, .975)
#     percapita.summary.stats.long[[j]]<-cbind(percapita.summary.stats.long[[j]],
#                                          as.factor(rep(c("lower_bound (2.5th percentile)", "media
#                                                        dim(percapita.summary.stats.long[[j]])[1]/
#     #strata.combos.summary.stats.wide[[j]]<-dcast(strata.combos.summary.stats.long[[j]], "Comorbid_co
#     percapita.summary.stats.wide[[j]]<-dcast(percapita.summary.stats.long[[j]], paste(paste(strata.co
#                                          value.var="V1")
#
#
#   }
#
#   return(list("summary.output" = strata.combos.summary.stats.wide,
#               "full.sims.output" = strata.combos.sims,
#               "summary.output.percapita"= percapita.summary.stats.wide,
#               "full.sims.output.percapita" = percapita.sims)
#   )
# }

setwd(paste(output_location, "/By_SubGroup", sep=""))
summ.stats.by.strata.cost<-summary.stats.by.strata(impact.sims.allgroups=impact.sims.total.allgroups,
                                                   strata.combos.list=strata.combos,
                                                   pop.sims=pop.sims)
summ.stats.by.strata.substitution.cost<-summary.stats.by.strata(impact.sims.allgroups=substitution.impa
                                                                strata.combos.list=strata.combos,
                                                                pop.sims=pop.sims)
summ.stats.by.strata.combined.cost<-summary.stats.by.strata(impact.sims.allgroups=combined.impact.sims.
                                                            strata.combos.list=strata.combos,
                                                            pop.sims=pop.sims)
summ.stats.by.strata.delta_forcost<-summary.stats.by.strata(impact.sims.allgroups=delta.sims.total.allg
                                                            strata.combos.list=strata.combos,
                                                            pop.sims=pop.sims)

summ.stats.by.strata.combined.consumed.cost<-summary.stats.by.strata(impact.sims.allgroups=combined.imp
                                                            strata.combos.list=strata.combos,
                                                            pop.sims=pop.sims)
summ.stats.by.strata.combined.unconsumed.cost<-summary.stats.by.strata(impact.sims.allgroups=combined.i
                                                            strata.combos.list=strata.combos,
                                                            pop.sims=pop.sims)

summ.stats.by.strata.current.intake.cost<-summary.stats.by.strata(impact.sims.allgroups=current.intake.
                                                                  strata.combos.list=strata.combos,
                                                                  pop.sims=pop.sims)
summ.stats.by.strata.CF.intake.cost<-summary.stats.by.strata(impact.sims.allgroups=CF.intake.impact.sim
                                                                     strata.combos.list=strata.combos
                                                                     pop.sims=pop.sims)
summ.stats.by.strata.current.intake.consumed.cost<-summary.stats.by.strata(impact.sims.allgroups=curren
```

```
                                                                  strata.combos.list=strata.combos,
                                                                  pop.sims=pop.sims)
summ.stats.by.strata.CF.intake.consumed.cost<-summary.stats.by.strata(impact.sims.allgroups=CF.intake.i
                                                             strata.combos.list=strata.combos,
                                                             pop.sims=pop.sims)
summ.stats.by.strata.current.intake.unconsumed.cost<-summary.stats.by.strata(impact.sims.allgroups=curr
                                                                          strata.combos.list=strata.co
                                                                          pop.sims=pop.sims)
summ.stats.by.strata.CF.intake.unconsumed.cost<-summary.stats.by.strata(impact.sims.allgroups=CF.intake
                                                                     strata.combos.list=strata.combos,
                                                                     pop.sims=pop.sims)

##inedible and wasted
summ.stats.by.strata.combined.inedible.cost<-summary.stats.by.strata(impact.sims.allgroups=combined.imp
                                                                    strata.combos.list=strata.combos,
                                                                    pop.sims=pop.sims)
summ.stats.by.strata.combined.wasted.cost<-summary.stats.by.strata(impact.sims.allgroups=combined.impac
                                                                      strata.combos.list=strata.combos
                                                                      pop.sims=pop.sims)

summ.stats.by.strata.current.intake.inedible.cost<-summary.stats.by.strata(impact.sims.allgroups=curren
                                                                          strata.combos.list=strata.co
                                                                          pop.sims=pop.sims)
summ.stats.by.strata.CF.intake.inedible.cost<-summary.stats.by.strata(impact.sims.allgroups=CF.intake.i
                                                                     strata.combos.list=strata.combos,
                                                                     pop.sims=pop.sims)
summ.stats.by.strata.current.intake.wasted.cost<-summary.stats.by.strata(impact.sims.allgroups=current.
                                                                            strata.combos.list=strata.
                                                                            pop.sims=pop.sims)
summ.stats.by.strata.CF.intake.wasted.cost<-summary.stats.by.strata(impact.sims.allgroups=CF.intake.imp
                                                                       strata.combos.list=strata.combo
                                                                       pop.sims=pop.sims)
```

Next, we're going to go through a for loop and output all outcomes into their own csv files for each strata combination (In my opinion, this is a better way to go about it than manually saving each file. Less prone to error, and robust to future changes).

First, start the for loop

```
for(j in 1:length(strata.combos))
{
```

In loop: First, save full sim outputs for total and per capita

```
setwd(paste(output_location, "/By_SubGroup/full_sims", sep=""))
#setwd("C:\\Users\\fcudhe01\\Box\\lasting_aim_3\\data\\out\\example_cost\\BySubGroup\\full_sims")
fwrite(x=summ.stats.by.strata.cost[[2]][[j]], file=paste("full.costenv.sims.output_by_", paste(strata
fwrite(x=summ.stats.by.strata.substitution.cost[[2]][[j]], file=paste("full.costenv.sims.substitution
fwrite(x=summ.stats.by.strata.combined.cost[[2]][[j]], file=paste("full.costenv.sims.combined.output_

setwd(paste(output_location, "/per_capita/By_SubGroup/full_sims", sep=""))
#setwd("C:\\Users\\fcudhe01\\Box\\lasting_aim_3\\data\\out\\example_cost\\per_capita\\BySubGroup\\ful
fwrite(x=summ.stats.by.strata.cost[[4]][[j]], file=paste("full.costenv.sims.percapita.output_by_", pa
fwrite(x=summ.stats.by.strata.substitution.cost[[4]][[j]], file=paste("full.costenv.sims.percapita.su
fwrite(x=summ.stats.by.strata.combined.cost[[4]][[j]], file=paste("full.costenv.sims.percapita.combin
```

Still in loop: Next, create outcome names to be used as variable names, and put relevant outcome data frames into two lists (for total and per capita). Within the k-loop, the summary stat variable names for the dataframes listed in outcome.data” are being changed so that they all consistently use the naming convention outlined in “outcome.names”.

```
#need to edit all this renaming code so it renames the new mean and SD columns. Perhaps take the oppu
## havinga set vector to rename, and using cat to combine the var name and the old name to create new

to.rename<-c("lower_bound (2.5th percentile)", "median", "upper_bound  (97.5th percentile)", "mean",
outcome.names<-c("impact",
                 "substitution_impact",
                 "combined_impact",
                 "delta",
                 "combined_consumed_impact",
                 "combined_unconsumed_impact",
                 "current_intake_impact",
                 "CF_intake_impact",
                 "current_intake_consumed_impact",
                 "CF_intake_consumed_impact",
                 "current_intake_unconsumed_impact",
                 "CF_intake_unconsumed_impact",
                 ##inedible, wasted
                 "combined_inedible_impact",
                 "combined_wasted_impact",
                 "current_intake_inedible_impact",
                 "CF_intake_inedible_impact",
                 "current_intake_wasted_impact",
                 "CF_intake_wasted_impact"
                 )
outcome.data<-list(summ.stats.by.strata.cost[[1]][[j]],
                   summ.stats.by.strata.substitution.cost[[1]][[j]],
                   summ.stats.by.strata.combined.cost[[1]][[j]],
                   summ.stats.by.strata.delta_forcost[[1]][[j]],
                   summ.stats.by.strata.combined.consumed.cost[[1]][[j]],
                   summ.stats.by.strata.combined.unconsumed.cost[[1]][[j]],
                   summ.stats.by.strata.current.intake.cost[[1]][[j]],
                   summ.stats.by.strata.CF.intake.cost[[1]][[j]],
                   summ.stats.by.strata.current.intake.consumed.cost[[1]][[j]],
                   summ.stats.by.strata.CF.intake.consumed.cost[[1]][[j]],
                   summ.stats.by.strata.current.intake.unconsumed.cost[[1]][[j]],
                   summ.stats.by.strata.CF.intake.unconsumed.cost[[1]][[j]],
                   ##inedible, wasted
                   summ.stats.by.strata.combined.inedible.cost[[1]][[j]],
                   summ.stats.by.strata.combined.wasted.cost[[1]][[j]],
                   summ.stats.by.strata.current.intake.inedible.cost[[1]][[j]],
                   summ.stats.by.strata.CF.intake.inedible.cost[[1]][[j]],
                   summ.stats.by.strata.current.intake.wasted.cost[[1]][[j]],
                   summ.stats.by.strata.CF.intake.wasted.cost[[1]][[j]]
                   )
outcome.data.per.capita<-list(summ.stats.by.strata.cost[[3]][[j]],
                   summ.stats.by.strata.substitution.cost[[3]][[j]],
                   summ.stats.by.strata.combined.cost[[3]][[j]],
                   summ.stats.by.strata.delta_forcost[[3]][[j]],
                   summ.stats.by.strata.combined.consumed.cost[[3]][[j]],
                   summ.stats.by.strata.combined.unconsumed.cost[[3]][[j]],
                   summ.stats.by.strata.current.intake.cost[[3]][[j]],
                   summ.stats.by.strata.CF.intake.cost[[3]][[j]],
                   summ.stats.by.strata.current.intake.consumed.cost[[3]][[j]],
                   summ.stats.by.strata.CF.intake.consumed.cost[[3]][[j]],
```

```
                    summ.stats.by.strata.current.intake.unconsumed.cost[[3]][[j]],
                    summ.stats.by.strata.CF.intake.unconsumed.cost[[3]][[j]],
                    ##inedible, wasted
                    summ.stats.by.strata.combined.inedible.cost[[3]][[j]],
                    summ.stats.by.strata.combined.wasted.cost[[3]][[j]],
                    summ.stats.by.strata.current.intake.inedible.cost[[3]][[j]],
                    summ.stats.by.strata.CF.intake.inedible.cost[[3]][[j]],
                    summ.stats.by.strata.current.intake.wasted.cost[[3]][[j]],
                    summ.stats.by.strata.CF.intake.wasted.cost[[3]][[j]]
)

for(k in 1:length(outcome.names))
{
 names(outcome.data[[k]])[which(names(outcome.data[[k]]) %in% c("lower_bound (2.5th percentile)", "me
   paste(outcome.names[k], c("lower_bound (2.5th percentile)", "median", "upper_bound  (97.5th percen
 names(outcome.data.per.capita[[k]])[which(names(outcome.data.per.capita[[k]]) %in% c("lower_bound (2
   paste(outcome.names[k], c("lower_bound (2.5th percentile)", "median", "upper_bound  (97.5th percen
}

# names(summ.stats.by.strata.cost[[1]][[j]])[which(names(summ.stats.by.strata.cost[[1]][[j]]) %in% c(
#   c("impact_lower_bound (2.5th percentile)", "impact_median", "impact_upper_bound  (97.5th percenti
# names(summ.stats.by.strata.substitution.cost[[1]][[j]])[which(names(summ.stats.by.strata.substituti
#   c("substitution_impact_lower_bound (2.5th percentile)", "substitution_impact_median", "substituti
# names(summ.stats.by.strata.combined.cost[[1]][[j]])[which(names(summ.stats.by.strata.combined.cost[
#   c("combined_impact_lower_bound (2.5th percentile)", "combined_impact_median", "combined_impact_up
# names(summ.stats.by.strata.delta_forcost[[1]][[j]])[which(names(summ.stats.by.strata.delta_forcost[
#   c("delta_lower_bound (2.5th percentile)", "delta_median", "delta_upper_bound  (97.5th percentile)
# names(summ.stats.by.strata.combined.consumed.cost[[1]][[j]])[which(names(summ.stats.by.strata.combi
#   c("combined_consumed_impact_lower_bound (2.5th percentile)", "combined_consumed_impact_median", "
# names(summ.stats.by.strata.combined.unconsumed.cost[[1]][[j]])[which(names(summ.stats.by.strata.com
#   c("combined_unconsumed_impact_lower_bound (2.5th percentile)", "combined_unconsumed_impact_median
# names(summ.stats.by.strata.current.intake.cost[[1]][[j]])[which(names(summ.stats.by.strata.current.
#   c("current_intake_impact_lower_bound (2.5th percentile)", "current_intake_impact_median", "curren
# names(summ.stats.by.strata.CF.intake.cost[[1]][[j]])[which(names(summ.stats.by.strata.CF.intake.cos
#   c("CF_intake_impact_lower_bound (2.5th percentile)", "CF_intake_impact_median", "CF_intake_impact_
# names(summ.stats.by.strata.current.intake.consumed.cost[[1]][[j]])[which(names(summ.stats.by.strata
#   c("current_intake_consumed_impact_lower_bound (2.5th percentile)", "current_intake_consumed_impac
# names(summ.stats.by.strata.CF.intake.consumed.cost[[1]][[j]])[which(names(summ.stats.by.strata.CF.i
#   c("CF_intake_consumed_impact_lower_bound (2.5th percentile)", "CF_intake_consumed_impact_median",
# names(summ.stats.by.strata.current.intake.unconsumed.cost[[1]][[j]])[which(names(summ.stats.by.stra
#   c("current_intake_unconsumed_impact_lower_bound (2.5th percentile)", "current_intake_unconsumed_i
# names(summ.stats.by.strata.CF.intake.unconsumed.cost[[1]][[j]])[which(names(summ.stats.by.strata.CF
#   c("CF_intake_unconsumed_impact_lower_bound (2.5th percentile)", "CF_intake_unconsumed_impact_medi
#
#
########################################
#
# names(summ.stats.by.strata.cost[[3]][[j]])[which(names(summ.stats.by.strata.cost[[3]][[j]]) %in% c(
#   c("impact_lower_bound (2.5th percentile)", "impact_median", "impact_upper_bound  (97.5th percenti
# names(summ.stats.by.strata.substitution.cost[[3]][[j]])[which(names(summ.stats.by.strata.substituti
#   c("substitution_impact_lower_bound (2.5th percentile)", "substitution_impact_median", "substituti
# names(summ.stats.by.strata.combined.cost[[3]][[j]])[which(names(summ.stats.by.strata.combined.cost[
#   c("combinedimpact_lower_bound (2.5th percentile)", "combined_impact_median", "combined_impact_upp
# names(summ.stats.by.strata.delta_forcost[[3]][[j]])[which(names(summ.stats.by.strata.delta_forcost[
#   c("delta_lower_bound (2.5th percentile)", "delta_median", "delta_upper_bound  (97.5th percentile)
```

```
#
# names(summ.stats.by.strata.combined.consumed.cost[[3]][[j]])[which(names(summ.stats.by.strata.combi
#   c("combined_consumed_impact_lower_bound (2.5th percentile)", "combined_consumed_impact_median", "
# names(summ.stats.by.strata.combined.unconsumed.cost[[3]][[j]])[which(names(summ.stats.by.strata.com
#   c("combined_unconsumed_impact_lower_bound (2.5th percentile)", "combined_unconsumed_impact_median
# names(summ.stats.by.strata.current.intake.cost[[3]][[j]])[which(names(summ.stats.by.strata.current.
#   c("current_intake_impact_lower_bound (2.5th percentile)", "current_intake_impact_median", "curren
# names(summ.stats.by.strata.CF.intake.cost[[3]][[j]])[which(names(summ.stats.by.strata.CF.intake.cos
#   c("CF_intake_impact_lower_bound (2.5th percentile)", "CF_intake_impact_median", "CF_intake_impact_
# names(summ.stats.by.strata.current.intake.consumed.cost[[3]][[j]])[which(names(summ.stats.by.strata
#   c("current_intake_consumed_impact_lower_bound (2.5th percentile)", "current_intake_consumed_impac
# names(summ.stats.by.strata.CF.intake.consumed.cost[[3]][[j]])[which(names(summ.stats.by.strata.CF.i
#   c("CF_intake_consumed_impact_lower_bound (2.5th percentile)", "CF_intake_consumed_impact_median",
# names(summ.stats.by.strata.current.intake.unconsumed.cost[[3]][[j]])[which(names(summ.stats.by.stra
#   c("current_intake_unconsumed_impact_lower_bound (2.5th percentile)", "current_intake_unconsumed_i
# names(summ.stats.by.strata.CF.intake.unconsumed.cost[[3]][[j]])[which(names(summ.stats.by.strata.CF
#   c("CF_intake_unconsumed_impact_lower_bound (2.5th percentile)", "CF_intake_unconsumed_impact_medi
# ```
```

Still in loop: Then, for both lists, merge their contents into a data frame.

```
list.names.intersect<-Reduce(intersect, lapply(outcome.data, colnames))

 all.summ.stats.by.strata<-Reduce(function(...) merge(..., by=list.names.intersect), outcome.data)
 all.summ.stats.by.strata.per.capita<-Reduce(function(...) merge(..., by=list.names.intersect), outcom
 #
 # all.summ.stats.by.strata<-Reduce(function(...) merge(..., all = T),list(summ.stats.by.strata.delta_
 #                                                                      summ.stats.by.strata.cost[[
 #                                                                      summ.stats.by.strata.substi
 #                                                                      summ.stats.by.strata.combin
 #
 #                                                                      summ.stats.by.strata.combin
 #                                                                      summ.stats.by.strata.combin
 #                                                                      summ.stats.by.strata.curren
 #                                                                      summ.stats.by.strata.CF.int
 #                                                                      summ.stats.by.strata.curren
 #                                                                      summ.stats.by.strata.CF.int
 #                                                                      summ.stats.by.strata.curren
 #                                                                      summ.stats.by.strata.CF.int
 #                                                                      )
 #                                                                      )
 #
 # all.summ.stats.by.strata.per.capita<-Reduce(function(...) merge(..., all = T),list(summ.stats.by.st
 #                                                                               summ.stats.by.st
 #                                                                               summ.stats.by.st
 #                                                                               summ.stats.by.st
 #
 #                                                                               summ.stats.by.st
 #                                                                               summ.stats.by.st
 #                                                                               summ.stats.by.st
 #                                                                               summ.stats.by.st
 #                                                                               summ.stats.by.st
 #                                                                               summ.stats.by.st
 #                                                                               summ.stats.by.st
 #                                                                               summ.stats.by.st
 #                                                                               )
 #                                                                               )
 #
```

Still in loop: Save the two outputs created (one for total and one for per capita) as csvs. Each of these dataframes will have summary stats for all outputs of interest for subgroup j.

```
 setwd(paste(output_location, "/By_SubGroup", sep=""))
 #setwd("C:\\Users\\fcudhe01\\Box\\lasting_aim_3\\data\\out\\example_cost\\BySubGroup")
 write.csv(x=all.summ.stats.by.strata, file=paste("summary.output_by_", paste(strata.combos[[j]], coll

 setwd(paste(output_location, "/per_capita/By_SubGroup", sep=""))
 #setwd("C:\\Users\\fcudhe01\\Box\\lasting_aim_3\\data\\out\\example_cost\\per_capita\\BySubGroup")
 write.csv(x=all.summ.stats.by.strata.per.capita, file=paste("summary.output.percapita_by_", paste(str
```

End loop.

```
}
```